

VNUHCM - University of Science
Faculty of Information Technology

CS486

Introduction to Database Systems

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Timetable

■ Lecture

- May 18 ~ Aug 11, 2026
 - Jun 08 ~ Jul 28 : 3 weeks off
 - Mon afternoon & Tue morning
- Office hour:
 - Sat, 13:00~16:00
 - Please make an appointment via email

■ Lab

- TAs
 - Nguyễn Ngọc Minh Châu, MSc. (nnmchau@fit.hcmus.edu.vn)
 - Nguyễn Ngọc Toàn, MSc. (nntoan@fit.hcmus.edu.vn)
- Time
 - Tue afternoon (I52)

Objective

- Describe
 - Roles of the database in an organization
 - Basic concepts of the database system
- Apply common data models in data modeling
- Master query languages to exploit the relational database
- Explain concepts of transactions, scheduler and locking, and recovery management
- Master
 - DB schema definition and query construction through AI-driven agents & DBMS tools
 - Critical thinking skills in practical contexts

Learning outcome

- (1) Interpret *roles of the database* in organizations
- (2) Describe *basic concepts* of the database
- (3) Draw *the entity relationship model (ER)* at a basic level
- (4) Interpret concepts of *the relational data model*
- (5) Master of *query languages*
- (6) Leverage AI agents and DBMS (*MS SQL Server*) capabilities to construct relational database schemas and retrieve data efficiently

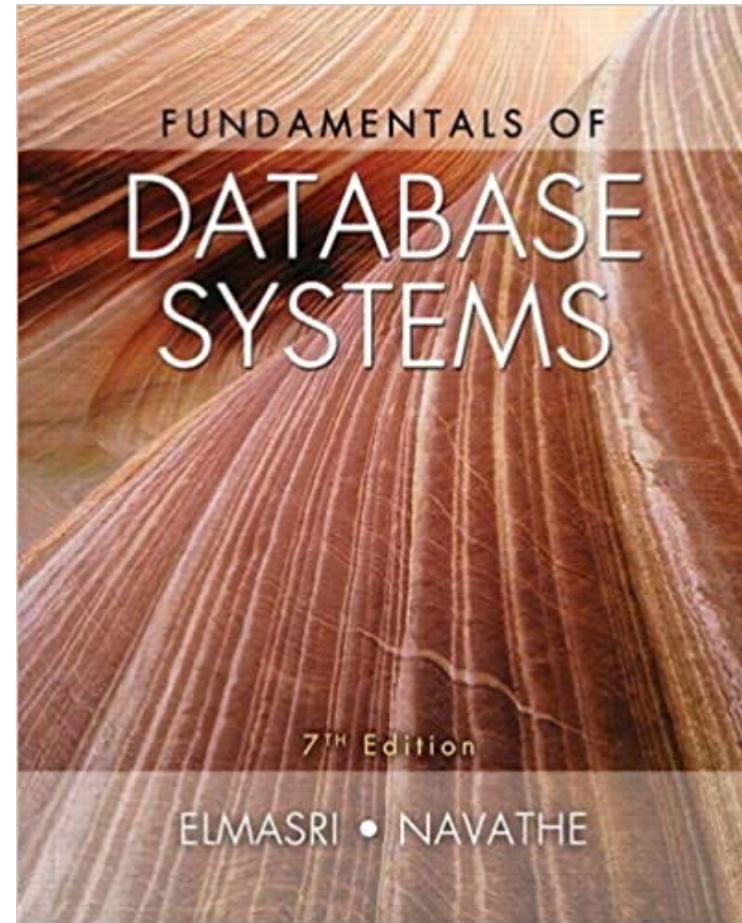
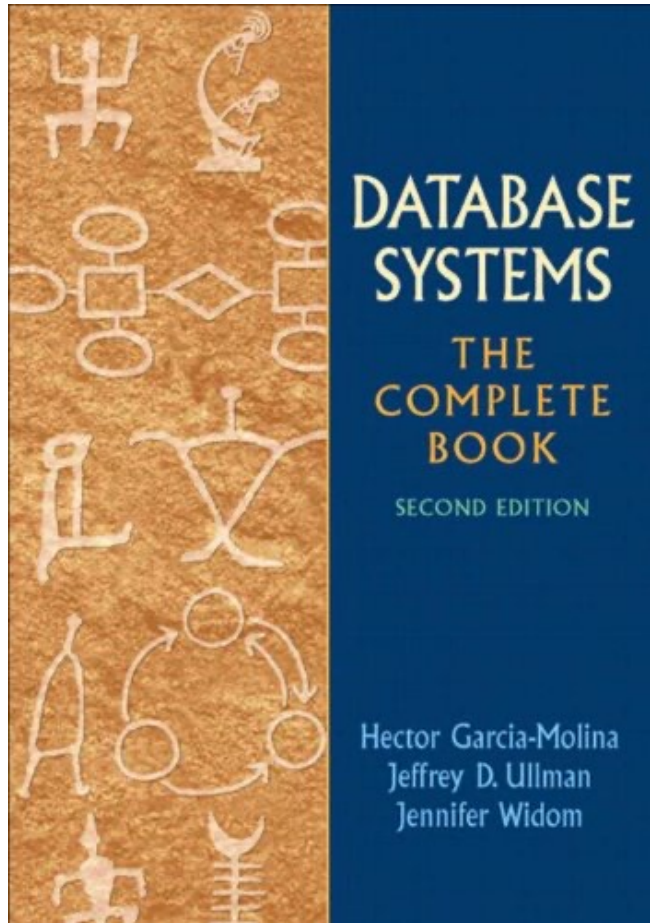
Learning outcome

- (7) Detect ***integrity constraints*** in the relational database
- (8) Categorize concepts of ***functional dependency and normal forms***
- (9) Explain how to ***transaction, scheduler, locking, and data recovery*** work
- (10) Describe ***future trends*** of the database

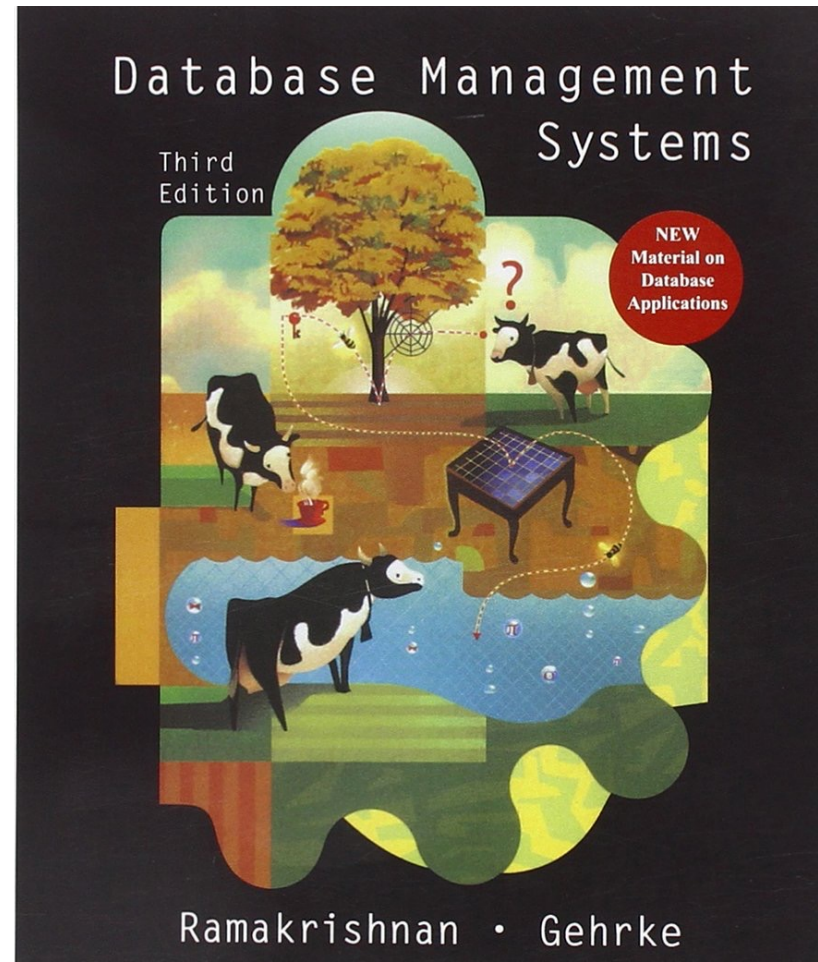
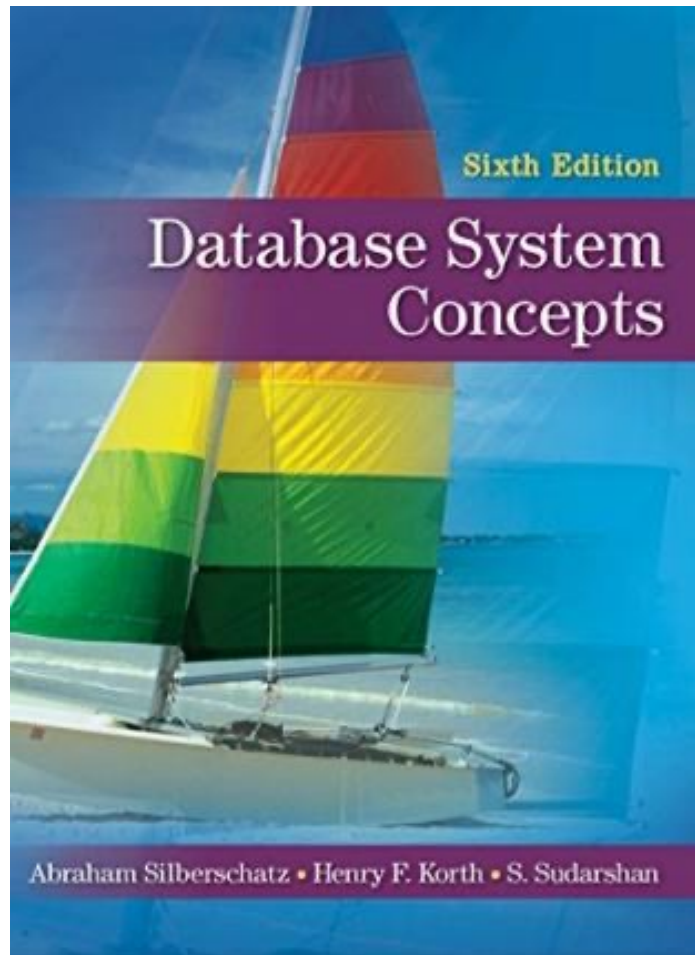
Content

- 01- Introduction
- 02- Entity relationship model
- 03- Relational data model
- 04- Query languages
- 05- Integrity constraint
- 06- Functional dependency and normal forms
- 07- Transaction and scheduler
- 08- Recovery management
- 09- Other issues

Textbook



Reference



Resource

- Moodle
 - Slide, video, assignment, lab document
 - Announcement, Q&A
- Google drive
 - @apcs.fitus.edu.vn

Grade

- **1. Lecture 40%**
 - Final exam: writing
 - Closed book, 100 minutes

- **2. Project 30%**
 - Proposal & final
 - Oral-based assessment

- **3. Lab (30%)**
 - Midterm & final exams
 - Closed book

Study method

- Flipped Classroom
 - 5 weeks
- At home (self study)
 - Watch video & read book
 - Take quizzes
 - Play with LLM models
- In class
 - Discussion
 - Q&A

Rule

- Project
 - Has to be done over 70%, grade will be counted
 - Is submitted late or does not follow the format → 0 point
- Cheating → fail
- If one of the three component scores is 0
 - The total grade will be 0

Rule (cont.)

- In class
 - Keep quiet (do not chat)
 - Switch your cell phone to silent mode
 - Raise your hand if you have comments/questions
 - Be on time
 - Do not bring food & beverage
 - Well dressed

- If you break the rules
 - Must quit the class
 - Be marked “absent” from class

Cheating: Description

- **Please pay close attention, especially if this is your first semester at CMU**
- What is cheating?
 - Sharing code: by copying, retyping, **looking at**, or supplying a file
 - Describing: verbal description of code from one person to another.
 - Coaching: helping your friend to write a lab, line by line
 - Searching the Web for solutions
 - Copying code from a previous course or online solution
 - You are only allowed to use code we supply, or from the CS:APP website
- What is NOT cheating?
 - Explaining how to use systems or tools
 - Helping others with high-level design issues
- See the course syllabus for details.
 - Ignorance is not an excuse

